The Gauss-Manin connection via the de Rham space

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Abstract. Let k be a field of characteristic 0 and $X \to S$ be a smooth morphism of smooth k-schemes. It is a classical result that the relative algebraic de Rham cohomology $\mathrm{H}^i_{\mathrm{dR}}(X/S)$ is naturally endowed with a flat connection, the *Gauss-Manin* connection. In this note, we provide an alternative perspective coming from the de Rham space. In particular, we explain how the relative de Rham cohomology (considered as a $\mathscr{D}_S$-module under the Gauss-Manin connection) is isomorphic to the higher direct image of $\mathscr{O}_{X_{\mathrm{dR}}}$ along the induced morphism between de Rham spaces $X_{\mathrm{dR}} \to S_{\mathrm{dR}}$.

Contents

1. The de Rham space

1.1. **Preliminaries.** We introduce the de Rham space, originally defined by Simpson [Sim96]. Many of the technical arguments in this section are found in [Rib24].

Let k be a fixed field of characteristic 0 and Aff_k be the category of affine k-schemes. Let τ be a Grothendieck topology on Aff_k. We define a τ-algebraic stack to be a sheaf $\mathrm{Aff}_k \to \mathrm{Grpd}$ in the τ topology. The category of τ algebraic stacks will be denoted AlgStk_τ.

Definition 1.1 (de Rham space)**.** Given any presheaf of $X : \mathrm{Aff}_k \to \mathrm{Set}$, we define X_{dR} to be the presheaf

$$X_{\mathrm{dR}}(A) := X(A^{\mathrm{red}})$$

Note that there is a natural map $X \to X_{\mathrm{dR}}$ and that associated to a morphism $X \to S$ there is an induced morphism $X_{\mathrm{dR}} \to S_{\mathrm{dR}}$. In this case, we define the relative de Rham space $(X/S)_{\mathrm{dR}}$ to be the fibered product $X_{\mathrm{dR}} \times_{S_{\mathrm{dR}}} S$.

Let X be a scheme which is formally smooth over k. If a k-algebra A has nilpotent nilradical (for instance, if A is Noetherian), then every A^{red}-point of X

extends to an A-point,

$$\begin{tikzcd} \operatorname{Spec} A^{\mathrm{red}} \arrow[r] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} A \arrow[r] \arrow[ur, dashed] & \operatorname{Spec} k \end{tikzcd}$$

by an infinitesimal lifting argument. Consequently, the natural map $X \to X_{\mathrm{dR}}$ is an epimorphism of presheaves. In general though, X smooth over k is sufficient.

Proposition 1.2. *If X is smooth over k, then $X \to X_{\mathrm{dR}}$ is an epimorphism.*

Proof. Let I_n be the ideal of A generated by elements of the form $x^n = 0$, and define $A_n := A/I_n$. Then $A^{\mathrm{red}} \cong \varinjlim_n A_n$, and

$$\operatorname{Hom}(\operatorname{Spec} A^{\mathrm{red}}, X) \cong \operatorname{Hom}(\varprojlim_n \operatorname{Spec} A_n, X) \cong \varinjlim_n \operatorname{Hom}(\operatorname{Spec} A_n, X)$$

where the second isomorphism holds because because X is locally of finite presentation [Sta18] Tag 01ZC. Consider an element on the right; this is given by a representative $\operatorname{Spec} A_n \to X$ for some n. By formal smoothness, this lifts to $\operatorname{Spec} A \to X$. □

From the above proof, we also have the following characterization: if X is locally of finite presentation over k, then

$$X_{\mathrm{dR}}(A) \cong \operatorname{Hom}(\varprojlim_n \operatorname{Spec} A_n, X) \cong \varinjlim_n \operatorname{Hom}(\operatorname{Spec} A_n, X)$$

and therefore X_{dR} is a colimit. An important consequence of this is the following.

Corollary 1.3. *If X is locally of finite presentation over k, $X_{\mathrm{dR}} : \mathrm{Aff}_k \to \mathrm{Set}$ commutes with finite limits and arbitrary colimits.*

An important property of the de Rham space is that it satisfies descent.

Proposition 1.4 ([Rib24], proposition 1.1.14)**.** *If X is a locally finite type scheme, the de Rham space X_{dR} satisfies fppf descent.*

We emphasize that k must have characteristic 0. Thus by our conventions, X_{dR} is an fppf algebraic stack.

1.2. **Algebraic de Rham cohomology.** A key feature of the de Rham space is that the cover $X \to X_{\mathrm{dR}}$ provides an identification of the algebraic de Rham cohomology of X with the $\mathscr{O}$-coherent cohomology of X_{dR}. We will provide a proof a relative version of this theorem.

Let $f : X \to S$ be a morphism of k-schemes, and consider the induced morphisms

$$\begin{tikzcd} (X/S)_{\mathrm{dR}} \arrow[r] \arrow[d, "\tilde{f}_{\mathrm{dR}}"'] & X_{\mathrm{dR}} \arrow[d, "f_{\mathrm{dR}}"] \\ S \arrow[r] & S_{\mathrm{dR}} \end{tikzcd}$$

Let $\mathscr{O}$ be the structure sheaf of $(X/S)_{\mathrm{dR}}$.

Theorem 1.5. *If $f : X \to S$ is a morphism of smooth k-schemes,*

$$\mathrm{H}^i_{\mathrm{dR}}(X/S) \cong R^i \widetilde{f}_{\mathrm{dR}\,*}\mathscr{O}$$

where the left is the Zariski hypercohomology of $\Omega^\bullet_{X/S}$.

To establish 1.5, we need the following (both of which are proved in [Rib24]).

Lemma 1.6. *Let X be locally of finite presentation. Then, the formal completion of X along a closed subscheme Z may be identified as*

$$\widehat{X}_Z \cong X \times_{X_{\mathrm{dR}}} Z_{\mathrm{dR}}$$

Lemma 1.7. *If X and S are locally of finite presentation,*

$$X \times_{(X/S)_{\mathrm{dR}}} X \cong (\widehat{X \times_S X})_\Delta$$

where the right is the formal completion of the image of the diagonal $X \to X \times_S X$.

Some remarks are in order. A priori, the definition of formal completion of X along a subscheme Z requires Z to be closed. However in 1.6, note that the right hand side does not require $Z \to X$ to be a closed immersion. Thus in 1.7, we do not require $X \to S$ to be separated and instead define formal completion as in 1.6.

Proof of 1.7. Let $Y = X \times_S X$, so that $\widehat{Y}_\Delta \cong Y \times_{Y_{\mathrm{dR}}} X_{\mathrm{dR}}$. The result follows from

$$\begin{tikzcd}
X \times_{(X/S)_{\mathrm{dR}}} X \arrow[r] \arrow[d] & Y = X \times_S X \arrow[d] \\
X_{\mathrm{dR}} \arrow[r, "\Delta"'] & Y_{\mathrm{dR}} \cong X_{\mathrm{dR}} \times_{S_{\mathrm{dR}}} X_{\mathrm{dR}}
\end{tikzcd}$$

being cartesian. $\square$

Proof of 1.5. For simplicity, we consider the absolute case $S = \operatorname{Spec} k$ first. Since $\widehat{X \times X} \rightrightarrows X \to X_{\mathrm{dR}}$ is a coequalizer of presheaves, $X \to X_{\mathrm{dR}}$ is an effective epimorphism. Therefore the simplicial scheme $\widehat{X}^\bullet = \check{\mathrm{C}}\mathrm{ech}(X \to X_{\mathrm{dR}})$ is a hypercover of X_{dR}. By 1.7, $\widehat{X}^n$ is the ind-scheme given by formal completion of the n-fold product $X \times \cdots \times X$ along the diagonal. By [Rod25], there is a quasi-isomorphism

$$\varprojlim \mathscr{O}_{\widehat{X}^\bullet} \to \Omega^\bullet_{X/k}$$

of Zariski sheaves on X, and consequently

$$R\Gamma(X_{\mathrm{dR}}, \mathscr{O}_{X_{\mathrm{dR}}}) \cong R\Gamma(\widehat{X}^\bullet, \mathscr{O}_{\widehat{X}^\bullet}) \cong R\Gamma(X, \Omega^\bullet_{X/k})$$

finishes the proof. The relative case $X \to S$ follows the same way. $\square$

1.3. **Quasi-coherent sheaves on de Rham spaces.** A key insight of Simpson was that for a smooth k-schemes X, that the category of quasi-coherent sheaves over X_{dR} identifies with the category of algebraic $\mathscr{D}_X$-modules on X. Since

$$\widehat{X \times X} \rightrightarrows X \to X_{\mathrm{dR}}$$

is a coequalizer of presheaves, we may consider X_{dR} as a quotient of X by the relation of *infinitesimal closeness* (see [Lur]). The data of a quasi-coherent sheaf on X_{dR} is thus equivalent to a quasi-coherent sheaf on X along with a compatible system of isomorphisms over points which lie in some n-fold formal thickening of the diagonal. This is equivalent to an action of the ring of differential operators on X.

Theorem 1.8. *If X is a smooth k-scheme, the category of quasi-coherent sheaves on X_{dR} is equivalent to the category of quasi-coherent $\mathscr{D}_X$-modules. That is,*

$$\mathrm{QCoh}(X_{\mathrm{dR}}) \cong \mathrm{QCoh}(\mathscr{D}_X)$$

See also [Rod25] §9. In particular, a locally free sheaf on X_{dR} is equivalent to a locally free sheaf along with a flat connection on X. Pulling back a locally free sheaf along $X \to X_{\mathrm{dR}}$ corresponds to forgetting the flat connection. Therefore, the fact that the relative de Rham cohomology sheaf $\mathrm{H}^i_{\mathrm{dR}}(X/S)$ may be endowed with a flat connection must correspond to the fact that it is pulled back from some quasi-coherent sheaf on S_{dR}.

Consider the following diagram,

$$\begin{tikzcd} (X/S)_{\mathrm{dR}} \arrow[r, "\widetilde{\pi}"] \arrow[d, "\widetilde{f}_{\mathrm{dR}}"'] & X_{\mathrm{dR}} \arrow[d, "f_{\mathrm{dR}}"] \\ S \arrow[r, "\pi"'] & S_{\mathrm{dR}} \end{tikzcd} \tag{$*$}$$

1.5 asserts that the relative de Rham cohomology is isomorphic to $R^i \widetilde{f}_{\mathrm{dR}\,*}\mathscr{O}$. Recall that there is a natural map

$$\pi^* R^i f_{\mathrm{dR}\,*}\mathscr{O}_{X_{\mathrm{dR}}} \to R^i \widetilde{f}_{\mathrm{dR}\,*}\widetilde{\pi}^*\mathscr{O}_{X_{\mathrm{dR}}}$$

called the base-change morphism. If this map is an isomorphism, observe that this would formally endow the relative de Rham cohomology with a flat connection: since $R^i f_{\mathrm{dR}\,*}\mathscr{O}_{X_{\mathrm{dR}}}$ is a quasi-coherent sheaf on S_{dR}, it is equipped with one by 1.8. The classical construction of the Gauss-Manin connection occurs because we were going "the wrong way" around the above cartesian square.

Proposition 1.9. *Let $f : X \to S$ be a smooth morphism of smooth k-schemes. If $(*)$ satisfies base-change, there exists a canonical flat connection ∇ on $\mathrm{H}^i_{\mathrm{dR}}(X/S)$.*

2. The six functor formalism for X_{dR}

Because of the non-algebraic nature of S_{dR}, establishing the above base change is slightly delicate; since S_{dR} is not a scheme, we cannot apply traditional proper or smooth base change for quasi-coherent sheaves on schemes. To this end, we proceed by setting up some abstract machinery: we appeal to the six functor formalism for quasi-coherent sheaves on X_{dR} as found in [Rib24] or [Rod25].

2.1. **Abstract 6-functor formalisms.** We recall the definition of a 6-functor formalism. Rather than strive for an exhaustive treatment, we will introduce only the minimal amount of theory to state the desired result. For details see [Sch22].

Definition 2.1. A *geometric set up* is a pair $(\mathcal{C}, E)$ where $\mathcal{C}$ is an ∞-category with finite limits and E is a collection of morphisms in $\mathcal{C}$ containing all isomorphisms and stable under pull back, composition, and taking diagonals.

Remark 2.2. E is the class of !-able morphisms, or morphisms which admit a $f_!$.

Definition 2.3. Given a geometric set up $(\mathcal{C}, E)$, the category of correspondences $\mathrm{Corr}(\mathcal{C}, E)$ is defined to be the category whose objects are the objects of $\mathcal{C}$ and a morphism from X to Y is a diagram

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ X & & Y \end{tikzcd}$$

in $\mathcal{C}$ where the arrow on the right $Z \to Y$ lies in E. A composition of morphisms is

$$\begin{tikzcd} & & Z \times_Y S \arrow[dl] \arrow[dr] & & \\ & Z \arrow[dl] \arrow[dr] & & S \arrow[dl] \arrow[dr] & \\ X & & Y & & T \end{tikzcd}$$

where the left arrow $Z \times_Y S \to T$ lies in E by assumption. There is a monoidal structure on $\mathrm{Corr}(\mathcal{C}, E)$ given by the product in $\mathcal{C}$.

Definition 2.4. Given a geometric set up $(\mathcal{C}, E)$, a 3-functor formalism is a lax symmetric monoidal functor $\mathcal{D} : \mathrm{Corr}(\mathcal{C}, E) \to \mathrm{Cat}_\infty$.

Given an arbitrary morphism $f : X \to Y$ in $\mathcal{C}$, the image of the correspondence

$$\begin{tikzcd} & X \arrow[dl, "f"'] \arrow[dr, equal] & \\ Y & & X \end{tikzcd}$$

is defined to be $f^* : \mathcal{D}(Y) \to \mathcal{D}(X)$. If $f : X \to Y$ lies in E, the image of

$$\begin{tikzcd} & X \arrow[dl, equal] \arrow[dr, "f"] & \\ X & & Y \end{tikzcd}$$

is defined to be $f_! : \mathcal{D}(X) \to \mathcal{D}(Y)$.

Definition 2.5. A 6-functor formalism is a 3-functor formalism

$$\mathcal{D} : \mathrm{Corr}(\mathcal{C}, E) \to \mathrm{Cat}_\infty$$

such that the following holds,

(a) the functors f^* and $f_!$ both admit right adjoints (defined to be f_* and $f^!$),

(b) for each object of $\mathcal{C}$, the symmetric monoidal ∞-category $\mathcal{D}(X)$ is closed.

A formal consequence of this is the following: given any Cartesian diagram

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

in $\mathcal{C}$ where f is !-able, there is a specified isomorphism of functors

$$g^* \circ f_! \to f'_! \circ (g')^*$$

[Man22]. This is a salient feature of 6-functor formalisms and the motivation for our consideration of them.

Definition 2.6. Let $f : X \to Y$ be !-able and $\omega_f := f^!(1_Y)$ be the dualizing object. The morphism f is said to be *cohomologically smooth* provided that

(a) The natural transformation $\omega_f \otimes f^*(-) \to f^!(-)$ is an equivalence,

(b) ω_f is $\otimes$-invertible in $\mathcal{D}(X)$,

(c) For any morphism $g : Y' \to Y$ with base change $f' : X' \to Y'$ of f, the above properties also hold for f'. Moreoever, if $g' : Y' \to Y$ denotes the base change of g, we also require the natural map

$$(g')^* \omega_f \to \omega_{f'}$$

to be an isomorphism.

For more details see [Sch22] §5. Cohomological smoothness is important because it yields abstract notion of smooth base change ([Zav23], lemma 2.3.9).

Theorem 2.7 (Cohomologically smooth base change)**.** *Let*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

be a Cartesian square in $\mathcal{C}$. Then the natural morphism

$$g^* \circ f_* \to (f')_* \circ (g')^*$$

is an isomorphism if g is cohomologically smooth.

As a final point, we mention briefly a particular type of cohomologically smooth morphism called cohomological étale morphisms. Intuitively, cohomological étale morphisms are cohomologically smooth with trivial dualizing object, i.e. satisfying $f^* \cong f^!$. We do not include a precise definition since cohomological smoothness is a sufficient condition for our needs (see [Zav23] for a precise definition).

2.2. **Quasi-coherent sheaves on algebraic stacks.** We outline a construction for a 6-functor formalism for the derived category of quasi-coherent sheaves on algebraic stacks. This is relies on extension theorems for 6-functor formalism which we will treat as a blackbox. This is done in full detail in [Rod25] and [Rib24].

As before, let Aff_k be the category of affine k-schemes.

Theorem 2.8. *The functor $\mathrm{D}_{\mathrm{qc}} : \mathrm{Aff}_k^{\mathrm{op}} \to \mathrm{Cat}_\infty$ which maps X to its derived ∞-category of quasi-coherent sheaves $\mathrm{D}_{\mathrm{qc}}(X)$ extends to a 6-functor formalism on the geometric setup $(\mathrm{Aff}_k, \mathrm{all})$ consisting of all morphisms.*

This is proved in [Sch22] §8.3. To extend this to a 6 functor formalism on algebraic stacks, we appeal to results of [HM24]. This uses the D_{qc}-topology, though we will not elaborate on this since we can restrict to the coarser fppf topology.

Theorem 2.9. *The six functor formalism on the geometric set up $(\mathrm{Aff}_k, \mathrm{all})$ extends uniquely to a six functor formalism $(\mathrm{AlgStk}_{\mathrm{D}_{\mathrm{qc}}}, E)$ such that*

(a) Let $f : Y \to X$ be a map whose pullback to every object in Aff_k lies in E, then f lies in E.

(b) Let $f : Y \to X$ be a map which is !-locally on the source or target in E; then f lies in E.

(c) Every map $f : Y \to X$ in E with $X \in Sch$ is !-locally on the source in E.

We emphasize that we do not have explicit knowledge of the class E. To check that a particular morphism of algebraic stacks is !-able, we need to verify one of the above conditions. For a detailed account, see [Sch22] *Appendix to Lecture IV: Passage to stacks* and [Rod25], Corollary 6.19.

Remark 2.10. The above result uses the D_{qc}-topology on algebraic stacks. However this does not pose a substantial problem; the D_{qc}-topology is *finer* than the fppf topology, and [Rib24] shows that there is a unique six functor formalism on fppf algebraic stacks whose !-able morphisms are precisely the morphisms whose sheafifications lie in the E above (proposition 1.3.23). That is, there is a compatible 6-functor formalism on the geometric set up $(\mathrm{AlgStk}_{\mathrm{fppf}}, E')$. In the sequel, we restrict to this formalism.

We need the following technical lemmas to proceed. Let X and Y be schemes.

Lemma 2.11 ([Sch22], Proposition 8.14)**.** *If $f : X \to Y$ is smooth and proper, then f is cohomologically smooth.*

Lemma 2.12 ([Rib24], Proposition 1.3.25)**.** *Let $Z \subset X$ be a closed subscheme, where the ideal sheaf is locally finitely generated. Then the morphism*

$$\widehat{X}_Z \to X$$

is cohomologically étale.

With these, we can prove the following central lemma.

Proposition 2.13. *If X is smooth over k, the natural morphism $X \to X_{\mathrm{dR}}$ is !-able and cohomologically smooth.*

We remark that the proof of this result is essentially ([Sch22], proposition 8.23), but Scholze works in the IndCoh-formalism.

Proof. Cohomological smoothness can be verified locally on the source and target. Since this is local on the source, we can restrict to an affine U. Moreover, by smoothness, we can assume U is of finitely presentation over k. By taking a compactification, we can restrict to a proper U. Let $Y \to U_{\mathrm{dR}}$ be a morphism with Y locally of finite presentation over k. Then the following diagram is Cartesian,

$$\begin{tikzcd}
(\widehat{U \times_k Y})_Z \arrow[r] \arrow[d] & Y \arrow[d] \\
U \arrow[r] & U_{\mathrm{dR}}
\end{tikzcd}$$

where $Z \subset U \times_k Y$ is the graph of $Y_{\mathrm{red}} \to U$, where Z is closed because U is separated and $Y^{\mathrm{red}} \to Y$ is a closed immersion. Then $(\widehat{U \times_k Y})_Z \to Y$ is the composite of $(\widehat{U \times_k Y})_Z \to U \times_k Y$ and $U \times_k Y \to Y$. Since the ideal sheaf of Z is finitely generated, the first morphism is cohomologically smooth by 2.12. The second is a base change of $U \to \operatorname{Spec} k$ and is thus cohomologically smooth by 2.11. $\square$

Proposition 2.14. *Let $f : X \to Y$ be a morphism between smooth, finite type k-schemes. Then $f_{\mathrm{dR}} : X_{\mathrm{dR}} \to Y_{\mathrm{dR}}$ is !-able in the 6-functor formalism for quasi-coherent sheaves.*

Proof. Since the class of !-able morphisms is stable under composition, it suffices to verify this after precomposing with the cohomologically smooth map $X \to X_{\mathrm{dR}}$. Then $X \to Y_{\mathrm{dR}}$ is the composite of f and $Y \to Y_{\mathrm{dR}}$, which are both !-able. $\square$

2.3. **The Gauss-Manin connection.** Let $f : X \to S$ be a morphism of smooth, finite type k-schemes and consider the diagram

$$\begin{tikzcd} (X/S)_{\mathrm{dR}} \arrow[r, "\widetilde{\pi}"] \arrow[d, "\widetilde{f}"'] & X_{\mathrm{dR}} \arrow[d, "f_{\mathrm{dR}}"] \\ S \arrow[r, "\pi"'] & S_{\mathrm{dR}} \end{tikzcd} \tag{$*$}$$

Then, 2.7 and 2.13 implies the following:

Corollary 2.15. *One has base-change for diagram* $(*)$*. In particular,*

$$\pi^* R^i f_{\mathrm{dR}*} \mathscr{O}_{X_{\mathrm{dR}}} \to R^i \widetilde{f}_* \widetilde{\pi}^* \mathscr{O}_{X_{\mathrm{dR}}}$$

is an isomorphism.

We emphasize the smoothness of f is unnecessary to establish base change for the relative de Rham square; since π is cohomologically smooth we need not assume any conditions on f_{dR} (other than that it's !-able). However, if f is not smooth, then $\mathrm{H}^i_{\mathrm{dR}}(X/S)$ will not be a locally free sheaf and thus we speak of $\mathscr{D}_S$-module structures instead of connections.

Corollary 2.16. *There is a canonical* $\mathscr{D}_S$*-module structure on the relative de Rham cohomology* $\mathrm{H}^i_{\mathrm{dR}}(X/S)$*.*

Moreover this $\mathscr{D}_S$-module is naturally isomorphic to the i-th higher direct image of the structure sheaf of X_{dR} along f_{dR}.

3. Appendix: generalities on cohomology theories

The Gauss-Manin connection on $\mathrm{H}^i_{\mathrm{dR}}(X/S)$ is a particular case of a general construction. Let k be a field of arbitrary characteristic, and τ a Grothendieck topology on Aff_k. We make the following nonstandard definition.

Definition 3.1. A $\diamondsuit$-cohomology theory is a cohomology theory for some class $\mathcal{S}$ of k-schemes with an associated six functor formalism $\mathrm{D}_\diamondsuit$ satisfying: For any $X \in \mathcal{S}$, there is a functorial assignment $X \rightsquigarrow X_\diamondsuit$, where the latter is a τ algebraic stack along with a morphism $X \to X_\diamondsuit$. This data is required to induce

(a) an isomorphism of cohomology,

$$H_\diamondsuit(X/k) \cong R\Gamma(X_\diamondsuit, \mathscr{O})$$

and likewise for relative cohomology with $(X/S)_\diamondsuit := X_\diamondsuit \times_{S_\diamondsuit} S$,

(b) and an equivalence of six functor formalisms,

$$\mathrm{D}_\diamondsuit(X) \cong \mathrm{D}_{\mathrm{qc}}(X_\diamondsuit)$$

In this context, we often call $\mathrm{D}_\diamondsuit$ the coefficients for the cohomology theory $H_\diamondsuit$.

Remark 3.2. de Rham, Dolbeault, crystalline, and prismatic cohomology are all examples of $\diamondsuit$-cohomology theories for some suitable class of k-schemes, with associated coefficients $\mathscr{D}$-modules, Higgs sheaves, crystals, and F-gauges, respectively. The assignment $X \rightsquigarrow X_\diamondsuit$ is sometimes called *transmutation* [Bha22]. The case of Betti cohomology is more subtle and requires the theory of analytic stacks [Sch24].

Let $f : X \to S$ be a morphism in $\mathcal{S}$ and consider the induced $f_\diamondsuit : X_\diamondsuit \to S_\diamondsuit$.

Proposition 3.3. *Suppose $X \to X_\diamondsuit$ is cohomologically smooth for D_{qc}. Then the relative cohomology $\mathrm{H}_\diamondsuit(X/S)$ has a canonical $\diamondsuit$-coefficient structure via $R(f_\diamondsuit)_*\mathscr{O}_{X_\diamondsuit}$.*

3.1. **The Kodaira-Spencer map.** Let k be a field of characteristic 0 and X a smooth k-scheme. Recall that the *algebraic Dolbeault cohomology* is defined to be

$$H^i_{\mathrm{Dol}}(X/k) := \bigoplus_{p+q=i} R^p f_* \Omega^q_{X/k}$$

Dolbeault cohomology is a $\diamondsuit$-cohomology theory in the sense of 3.1 [Sim99]: there exists a fppf algebraic stack X_{Dol} and a cover $X \to X_{\mathrm{Dol}}$ whose $\mathscr{O}$-coherent cohomology recovers H_{Dol}. The category of quasi-coherent sheaves on X_{Dol} is equivalent to the category of *Higgs* sheaves on X. The stack X_{Dol} has a simplicial presentation as $\widehat{T^\bullet X}$, where $T^n X$ is the n-fold product $TX \times_X \cdots \times_X TX$ and the formal completion is along the zero section $X \to T^n X$. By the method of 2.13,

Corollary 3.4. *If X is smooth over k, $X \to X_{\mathrm{Dol}}$ is cohomologically smooth. Consequently, there is a natural Higgs field ϑ on $H^i_{\mathrm{Dol}}(X/S)$ and*

$$(H^i_{\mathrm{Dol}}(X/S), \vartheta) \cong R^i f_{\mathrm{Dol}\,*} \mathscr{O}_{X_{\mathrm{Dol}}}$$

If we assume $f : X \to S$ is a smooth morphism, this is precisely the Higgs structure induced via the Kodaira-Spencer map $\kappa : \Omega_S^\vee \to R^1 f_* \Omega^\vee_{X/S}$. Cup product with the associated section in $R^1 f_* \Omega^\vee_{X/S} \otimes \Omega_S$ gives an $\mathscr{O}_S$-linear map

$$R^p f_* \Omega^q_{X/S} \to R^{p+1} f_* \Omega^{q-1}_{X/S} \otimes \Omega_S$$

This yields an $\mathscr{O}_S$-linear map $\vartheta : H^i_{\mathrm{Dol}}(X/S) \to H^i_{\mathrm{Dol}}(X/S) \otimes \Omega_S$ with $\vartheta \wedge \vartheta = 0$.

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